

GeoNLI : Natural Language Interpretation of Satellite Imagery

Team 18

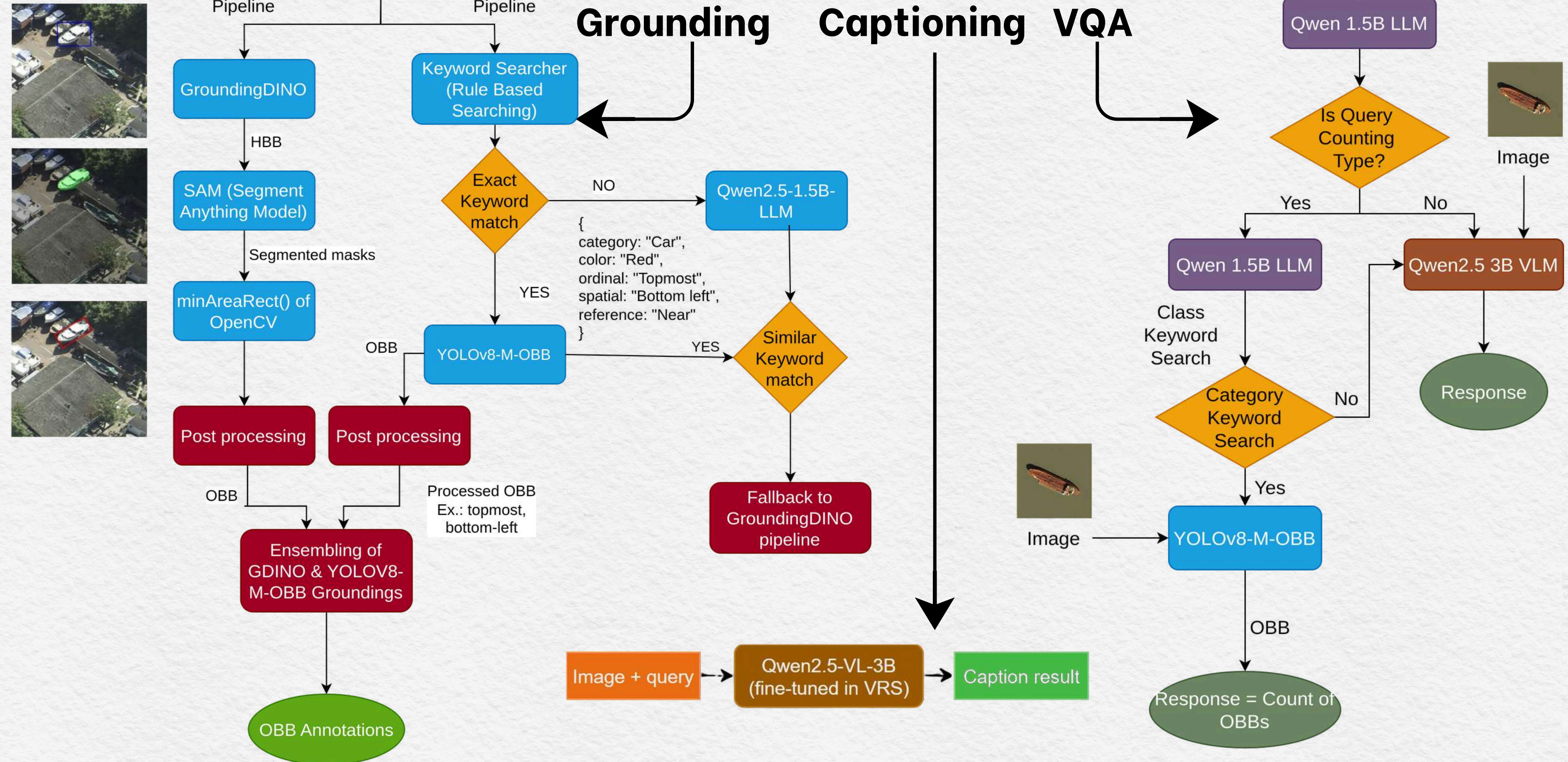


Solution Design

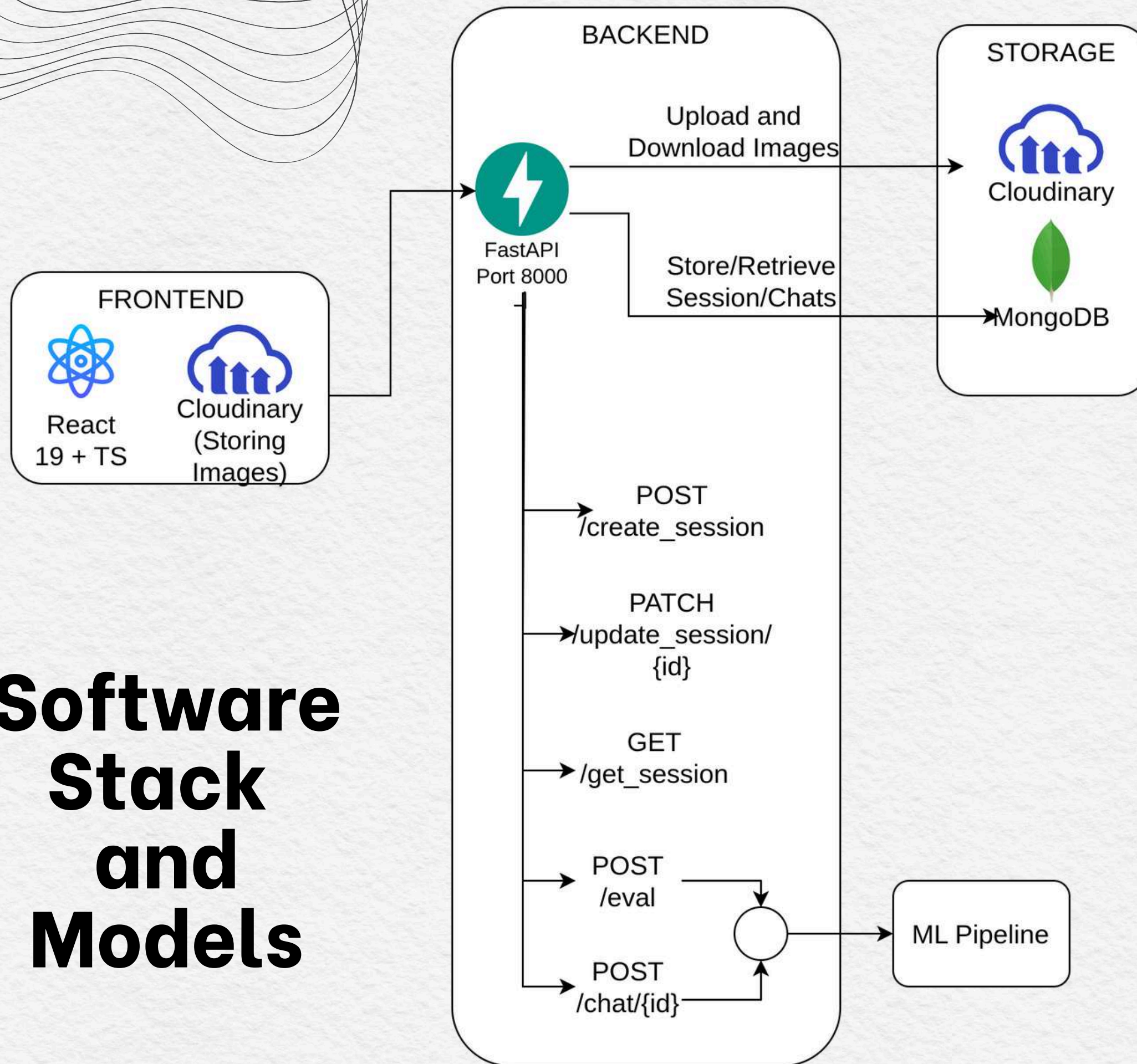
Grounding

Captioning

VQA



Software Stack and Models



Backend

- Language: Python 3.10
- Framework: FastAPI (v0.115.0)
- Database: MongoDB

Frontend

- Language: TypeScript (~5.9.3)
- Framework: React (v19.2.0)

Models

1.) Captioning:

- Qwen2.5-VL-3B Fine-tuned

2.) Grounding:

- Grounding-DINO-base + SAM 1 ViT base (Pipeline 1)
- Qwen 2.5-1.5B + YOLOv8-M-OBB (Pipeline 2)

3.) VQA:

- Qwen2.5 1.5B LLM
- Qwen2.5-VL-3B Fine-tuned
- YOLOv8-M-OBB Fine tuned

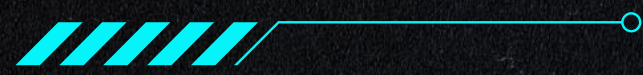
Innovations

Limitations of Existing Grounding Solutions

- **GroundingDINO**: Lacks satellite-image context and outputs only HBB not OBB
- **GeoChat / Qwen2.5-VL**: Language decoder is **not designed** for **precise numeric outputs**, making bounding-box **coordinate prediction unreliable**.
- **VQA Counting by VLMs**: **Language head** predicts numbers poorly; cross-attention isn't optimized for object counting.

Our VQA Counting Innovation

- Use **Qwen2.5-1.5B LLM** to identify if the query requires counting.
- A **secondary pass from LLM** maps the query to the most **relevant YOLO class**.
- Route the image through **YOLO-OBB** for **precise, oriented object detection**.
- Compute the **count** directly from the **number of detected OBBs**, bypassing **unreliable** language-based numeric prediction.



Innovations

OBB Consensus Fusion: Multi-Pipeline Grounding Innovation



Query Parsing & Rule-Based Filtering

- Convert natural language prompts into a structured query JSON, filter bounding boxes by color, spatial location, and other attributes.

Pipeline 1 – GroundingDINO + SAM

- Generated HBBs with GroundingDINO + SAM Pipeline and filtered using query-based color and positional attributes.

Pipeline 2 – Finetuned YOLO for Satellite Images

- The query JSON parsed to select YOLO classes, OBBs predicted and filtered by class, color, and spatial constraints, with fallback to Pipeline 1 if no class was parsed.

Consensus OBB Fusion (Cross-Model Agreement)

- Confidence-Weighted OBB Ensemble: Merges detections from both pipelines.
- **Spatial Overlap Weighting:** Overlapping boxes with high IoU are fused into a single, precise OBB.
- **Confidence Averaging:** Prediction scores from both models are leveraged to refine final box positions.
- **Redundant-Box Suppression:** Eliminates duplicates while preserving unique detections.



Testing Strategy & Results

Similarity-Aware Sampling

- **Similarity-Aware Sampling:** Evaluate every k-th image to cut computation while preserving representative VQA and captioning accuracy.
- **Adaptive Category Optimization (ACO):** We apply targeted improvements specifically to the categories where performance is weak.

Grounding Results

* on Annotations Val VRS

	Recall	S_grounding
Dino + SAM	0.3087	0.142
YOLO	0.5095	0.364
Combined	0.6068	0.364

VQA Results

* on 1500 Images VRS

Model	Binary	Numeric	Semantic
Baseline Qwen 3B	0.7978	0.4734	0.8475
Fine-Tuned Model	0.8407	0.5468	0.9062
Qwen + YOLO	-	0.6184	-

Captioning Results

Baseline Qwen 3B	Fine-Tuned Model
0.6313	0.7375



Limitations & Future Scope

RESTRICTED CATEGORY COVERAGE IN YOLO-OBB

- The **YOLO-OBB** model operates on a **fixed, closed set of categories**, limiting its ability to detect **objects outside its trained classes**.
- The **supported category set** can be **expanded** by **training** the model on datasets that include a **larger and more diverse range of classes**.

REMOTECLIP-YOLO INTEGRATED SPATIAL REASONING

- Use **RemoteCLIP's 3×3 region-based** confidence to guide **Qwen's reasoning**.
- Select the best output between **RemoteCLIP** and **YOLO** in **low-confidence cases**.
- Improves **positional understanding** and **open-vocabulary object handling**.



Thank You

